MILITARY SPECIFICATION

SPITBACK LEAD ASSEMBLY FOR FUZE, POINT DETONATING, M759

This specification is approved for use by the US Army Armament Munitions, and Chemical Command and is available for use for all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains the requirements and quality assurance provisions for fabrication of parts, assembly and packing for a spitback lead assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitations, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-Q-9858	-	Quality Program Requirements

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1167	-	Ammunition Data Card

FSC: 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings and publications. The following Government documents drawings, and publications form a part of this specification to the extent specified herein:

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER

PRODUCT AND PACKING DRAWINGS

11834312	-	Fuze, Pd, M759
11834335	-	Spitback Lead Assembly
11834340	-	Detonator and Lead Propagation Test Assembly
11834346		Spitback Propagation Test Assembly

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activcity or as directed by the Contracting Officer.)

PUBLICATIONS

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Officer, Washington, DC 20402. Orders should specify, "49 CFR100-199 (latest revision)")

2.2 Order of precedence. In the event of a conflict between this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Manufacturing process. All parts and assemblies shall be manufactured by the equivalent process used for first article approval and no deviations from that process shall be made without prior contracting officer approval. (See 4.5.1).

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3.2 First article inspection. This specification contains provisions for first article inspection. Requirements for the submission of first article sample of the contractor shall be as specified in the co ntract.

3.3 Functioning.

3.3.1 Sensitivity and output. The spitback lead assembly when initiated by a high order detonator output will function high order and will indent the steel witness block to a depth of not less than 0.025 inches when tested as specified in 4.5.2.1.

3.3.2 Varicomp initiation. The spitback lead assembly when initiated by a higher order detonator output will function high order and cause a 21.49% Varicomp explosive (NOL PR 6391 mix X-N0385) at 1.92 inch minimum air gap to function high order and indent a steel witness block to a depth of not less than 0.015 inch when tested as specified in 4.5.2.2

3.4 Workmanship. All parts and assemblies shall be fabricated finished in thorough, workmanlike manner. All parts and assemblies shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly, nor shall they be contaminated by the cleaning agent. All required markings and stampings (for packing) shall be neat and sharply defined. Required packing materiels shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the following items in sample quantities as indicated.

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PART DESCRIPTION	DRAWING	QUANTITY
Spitback Lead Assemblies	11834335	336

4.3.2 Inspection to be performed. See MIL-A-48078 and Table 1 included herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

MIL-S-70437(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Spitback Lead Assembly	1 OF 2	11834335
			NEXT HIGHER ASSEMBLY: See paragraph 6.2

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
	Load Assembly				
	Examination for defects	315	Note 1	3.2,3.3	4.4.2.1 and 4.4.2.2
	Functioning - sensitivity and output	315	Note 1	3.5.1	4.5.2.1
	Functioning - varicomp initiation	21	Note3,4	3.5.2	4.5.2.2

NOTE 1 Accept with three or less defects, reject on four or more defects.

NOTE 2 Accept with two or less defects, reject on three or more defects.

NOTE 3 Demonstrate a 99.99% reliability at 95% confidence using the following accept/reject criteria:

for n = 4, a = 0, r = 9 (minimum sample)
n = 7, a = 1, r = 9
n = 9, a = 2, r = 9
n = 11, a = 3, r = 9
n = 13, a = 4, r = 9
n = 15, a = 5, r = 9
n = 17, a = 6, r = 9
n = 19, a = 7, r = 9
n = 21, a = 8, r = 9 (minimum sample)

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70437(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Spitback Lead Assembly				11834335
					NEXT HIGHER ASSEMBLY
					See paragraph 6.2
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
NOTE 4	Varicomp is a method for evaluation the interface between explosive components in which it can be determined by statistical analysis and testing that the reliability of a fuze explosive train can be predicted at high confidence levels with a small number of tests. This is done by varying the sensitivity of different explosives that are then substituted into the explosive train to determine the reliability limits under a penalty test situation. For reliability tests the acceptor design explosive is replaced by a less sensitive "varicomp" explosive. (Reference NWC TP 5789 12/75)				

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Each lot shall contain PBXN-5 from one manufacturer with one lot interfix number.

4.4.2 Examination. (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance wtith 4.4.4.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	PBXN-5	1 OF 1	11834335
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Output charge properly sieved		each lot	3.2	Certification
MINOR:	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Spitback Lead Assembly (Non-destructive Tests)	1 OF 1	11834335
			NEXT HIGHER ASSEMBLY: See paragraph 6.1

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Total height		0.40%	3.2	Gage
102.	Main body height		0.40%	3.2	Gage
103.	Outside diameter (small)		0.40%	3.2	Gage
104.	Outside diameter (large)		0.40%	3.2	Gage
105.	Caked explosive on exterior		0.40%	3.2	Visual
106.	Depth of output dimple		0.40%	3.2	Gage
107.	Cracks or splits		100%	3.2	Visual
MINOR:					
201.	Selant less than 360° or wrong color		0.65%	3.2,3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 11834335
4.4.2.3	Spitback Lead Assembly (Destructive Tests)				NEXT HIGHER ASSEMBLY See paragraph 6.1
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Sensitivity and output	21	0.40%	3.5.1	4.5.2.1
102.	Varicomp initiation		Note 1,2	3.5.2	4.5.2.2
MINOR:	None defined				
NOTE 1	Demonstrate a 99.99% reliability at 95% conficence using the following accept/reject criteria: for n = 4, a = 0, r = 9 (minimum sample) n = 7, a = 1, r = 9 n = 9, a = 2, r = 9 n = 11, a = 3, r = 9 n = 13, a = 4, r = 9 n = 15, a = 5, r = 9 n = 17, a = 6, r = 9 n = 19, a = 7, r = 9 n = 21, a = 8, r = 9 (minimum sample)				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.3	Spitback Lead Assembly (Destructive Tests)				11834335
					NEXT HIGHER ASSEMBLY
					See paragraph 6.1
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
NOTE 4	Varicomp is a method for evaluation the interface between explosive components in which it can be determined by statistical analysis and testing that the reliability of a fuze explosive train can be predicted at high confidence levels with a small number of tests. This is done by varying the sensitivity of different explosives that are then substituted into the explosive train to determine the reliability limits under a penalty test situation. For reliability tests the acceptor design explosive is replaced by a less sensitive "varicomp" explosive. (Reference NWC TP 5789 12/75)				

NOTES:

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4.4.3 Testing. Testing is described in the first article table and quality conformance inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the "Paragraph Reference/Inspection Method" column starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6.2 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.

PRECAUTION - This specification covers sampling and testing of explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Section 3 and 5 of this specification in accordance with 4.4.2 herein and Table 1.

4.5.2 Functioning.

4.5.2.1 Sensivitity and output. The sample spitback lead assemblies shall be randomly selected from each lot. The lead shall be assembled in accordance with drawing 11834340. A weight of 0.25 ounce plus or minus 0.01 ounce shall be dropped on the firing pin, to function the detonator, from a height of five (5) inches minimum. A lead that fails to fire or fires but fails to produce a dent in the steel witness block of the required depth shall be classed defect if the detonator confinement sleeve expansion is 0.330 inch diameter or greater. If the confinement sleeve expansion is less than 0.330 inch diameter, the lead shall not be classed defective, but the detonator classed defective and the result of a no-test. The indentation depth shall be measured using a probe with a 0.017 inch plus or minus 0.002 inch spherical radius tip.

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4.5.2.2 Varicomp initiation. The sample spitback lead assemblies shall be randomly selected from each lot. The lead shall be assembled in accordance with drawing 11834346 and functioned as in 4.5.2.1. A lead that fails as specified in 4.5.2.1 shall be classed defective. The indentation depth shall be measured using the probe specified in 4.5.2.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, marking and labeling requirements. Packaging, packing, marking and labeling shall meet the Department of Transporation Regulation 49 CFR 100-199 and shall be submitted to and approved by the contracting officer prior to the initial shipment. No subsequent changes shall be made prior to obtaining new approval. Part number, lot number and description shall be clearly marked on the top of each inner box anc carton. The inner packing must confine the lead during shipment. The inner package shall permit quick and easy removal of the lead by the user without the packaging material contaminating the product. the outer packing must contain lot number, part number, nomenclature and hazard marking information.

6. NOTES

6.1 Intended use. The spitback lead assemblies are intended for use in the following fuze -

30MM M759 11834312

6.2 Order data. Procurement documents should specifiy the following:

6.2.1 Title, number and date of this specification.

6.2.2 Place of inspection if not at the place of manufacture.

6.2.3 Provision for the supply, maintenance, and disposition of mandatory test equipment for acceptance inspection process.

6.2.4 Instructions for shipment of production lot samples for functional tests.

6.2.5 Provision for preparation for delivery if required for indefinite storage of inter-plant shipment of the lead assemblies.

6.2.6 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-Q-9858, Quality Program Requirements.

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6.2.7 Provisions for MIL-STD-1167, Ammunition Data Cards, on DD Form 1423, Contract Data Requirements List.

6.3 Submission of inspection equipment designs for approval. (See MIL-A-48078). Submit two (2) copies of designs as required to: AMCCOM, ATTN: DRSMC-QAF-S (D), Dover, NJ 07801.

Custodian:
Army-AR

Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-70437(AR)	SPITBACK LEAD ASSEMBLY FOR FUZE, POINT DETONATING, M759

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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